Praktikum 1 Report

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Wien, 2024

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Contents

Abstract

This report aims at providing clarity about how the previous bachelor thesis project was extended with new capabilities. The upgrade to the old project includes automatic speech recognition, two types of summarization, topic modelling and visualisation of the output with word cloud, NER heatmap and POS sunburst chart. Since the previous project used the Flask framework, this extension uses Flask as well. The aim was to further improve the overall analysis by using different LLMs, like multilingual T5 for abstractive summarization or Whisper, a general-purpose speech recognition model. Furthermore, Latent Dirichlet Allocation (LDA), which is a probabilistic generative model, was used for topic modelling and Term Frequency - Inverse Document Frequency (IF-IDF) was used for extractive summarization. The output of the program was extended with the previously mentioned graphs in order to bridge the gap between human understanding and computational results, improving the rate of spotting patterns, errors and distributions in the output and ultimately evaluating the performance of the code.

1 Introduction

Automatic text processing and analysis offer many opportunities across multiple fields such as education, digital humanities, accessibility and information retrieval[12]. The ability to analyze, visualize and transform the natural language input allows for the extraction of meaningful data for certain tasks. While such techniques are well developed for English, many other languages remain underrepresented, such as Serbian, since it has a smaller pool of speakers and more complex morphology, relatively free word order, two different scripts, and multiple dialects[4]. All of this makes the Serbian language an important case for testing and improving the existing NLP tools.

The first version of the application, which was completed during the bachelor's thesis, included morphological, semantic and syntactic analysis, offering the user tokenization, lemmatization, part-of-speech tagging, named entity recognition, dependency parsing, translation into English, transliteration between Latin and Cyrillic, linking local and online definition of the words and visualization of the dependency tree. The new extension to this base includes new input modalities, discourse-level analyses and new visualisations for interpretability.

At the center of the new additions lies the integration of the speech input. The user can now choose among multiple input modes. Apart from the keyboard input, the user can record their own speech or upload a sound file. The input sound will then be processed using OpenAI's medium Whisper model which then transcribes the sound into text and inserts it into the input field, starting the analysis[19]. This represents a big step towards accessibility of the application.

Apart from the new input, the system now has two additional approaches to summarization implemented: extractive summarization, which identifies and returns the most important sentences using the graph of sentence similarities[13] and abstractive summarization, which uses the multilingual transformer model to generate a shortened version of the input [6]. This upgrade serves as an entry point to pragmatic and discourse level analysis in the application, allowing the users to grasp the essence of the text in optimal amount of time.

A second discourse-level addition is topic modelling, that uses Latent Dirichlet Allocation, which is a probabilistic model that discovers latent themes across documents[1]. What that means is that the model identifies groups of words that frequently co-occur.

The final extension to the application is the new graphs, which aim at improving the understanding of the users. This includes a word cloud that highlights the most frequent words[7], an NER heatmap that shows the distribution of named entities across the text and finally a POS sunburst chart that shows the syntactic makeup of the corpus[9].

2 Automatic Speech Recognition (ASR)

Automatic Speech Recognition (ASR) is a method of translating spoken language into written text and serves as a base for Natural Language Processing. It is applied in multiple areas like accessibility, transcription, language learning and human-computer interaction. Traditional ASR systems have relied on acoustic models, pronunciation dictionaries and statistical language models[18], however, with the rise of deep learning, end-to-end neural architectures have become the standard in the new systems, significantly improving the accuracy[8].

The advances in the deep learning made the training of the end-to-end ASR models possible. Such models have their entire system represented as a single neural network. Connectionist Temporal Classification (CTC), Transformer-based models and sequence-to-sequence models with attention have replaced the older modular designs. In such systems, raw audio features are directly mapped to text, cutting down the need for manually designed components and improving the generalization of the models across the domains[5][2][3].

However, even though the ASR has improved for many high-resource languages, many low-resource languages still remain underrepresented. Building a robust ASR system requires large annotated speech corpora, which are not always found or are lacking for less popular languages like Serbian. The solution to this is the usage of multilingual training, which jointly trains the data from many languages. During the training, the models can learn the acoustic and linguistic patterns that transfer across the languages. This approach to training is especially good for morphologically low or low-resource languages, where the necessary datasets are scarce[17].

In this project, the implementation of the ASR system was realized using the OpenAI's medium Whisper model. This model is a transformer-based model, that was trained on 680,000 hours of multilingual and multitask supervised data collected from the web. It is also important to mention that Whisper was also trained for translation and language identification. Whisper, of course, supports Serbian transcription, which was made possible because of the multilingual training, allowing Whisper to generalize the knowledge of the high-resource language similar to Serbian[19].

The below image shows the newly updated input bar of the application. This version has two additional buttons in the design. The first button, which uses the microphone emoji is used for directly inputting user voice into the system, which is then processed by the Whisper. The processed text is then input in the bar and the analysis begins automatically. The second option is to upload an audio file to the application directly, which then goes through the same procedure.

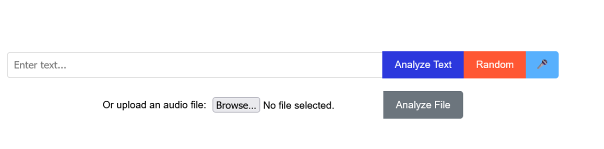

Figure 1: The new input field of the application.

The performance of the medium model is fairly robust and was tested using the JuzneVesti-SR v1.0 dataset, which consists of audio recordings with a total length of 50.55 hours, divided into 10,811 entries, where each entry is represented as a WAV, PCM, mono file, sampled at 16 kHz. Each entry ranges from 2 to 30 seconds. The dataset was compiled from the talk show "15 minuta", produced by the Južne Vesti news outlet[21]. Of the 10,811 entries, 1000 were taken for the performance analysis because of the hardware limitations. The scatter plot below shows the relationship between word error rate (WER) and character error rate (CER). The CER values are lower than WER values in most cases, which was expected. This means that Whisper produces phonetically and orthographically similar outputs that still diverge at the word level. The majority of files fall within 20-60% WER band and 0-40% CER band. The average WER was 45.92% and for the CER it was 21.46%. While Whisper often reports low WER on high-resource languages [19], Serbian remains more challenging. This reflects the increasing complexity of Serbian language combined with the scarcity of training resources. The most frequent errors observed were caused by the inability of the model to detect the right dialect, repetitions of the words and deletions of short function words. In addition to that, some names were not correctly transcribed, too. The duration of the audio clip did not appear to correlate with the error rates, indicating that the errors are more likely due to linguistic or acoustic properties.

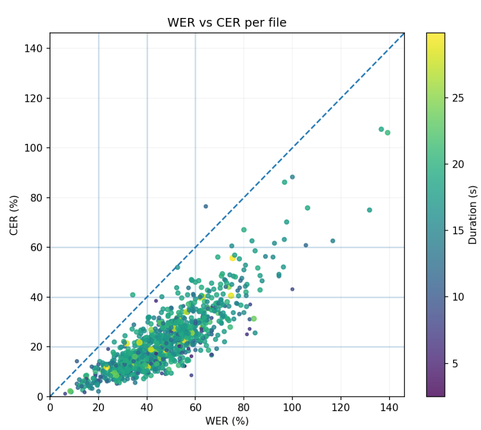

Figure 2: WER vs CER per file

3 Summarization

The summarization is the process of shortening the original text while still preserving the original context and coherence of the text[10]. It is widely used for aggregating the news, academic research, education and retrieval of information since it allows the user to quickly grasp the essence of the larger texts in less time. There are two main types of summarization: *extractive* and *abstractive*. Extractive methods shorten the text by choosing the most important sentences or phrases from the input text and discard the rest, while abstractive metods generate new sentences that keep the core idea but use less words in order to do so[14]. This project integrates both of these types, providing multiple ways of analyzing and interpreting Serbian texts, which represents an expansion towards discourse-level analysis.

3.1 Extractive Summarization

As mentioned before, extractive summarization is an approach where a summary is created by selecting only the most important sentences from the original text. This method assumes that some sentences carry more significant data than the others and aims to identify and extract them. This method is language-independent and does not need annotated training data, making it a perfect fit for low-resource languages such as Serbian[14]. The extractive summary in this project was implemented by using a graph-based algorithm similar to TextRank[13]. The whole process begins by segmenting the input text into separate sentences, which are then transformed into a numerical representation using the Term Frequency - Inverse Document Frequency (TF-IDF) weighting scheme. TF-IDF emphasizes words that are frequent in the sentence but relatively rare in the whole document. This helps highlight the distinctive content of each sentence[11]. A fully connected similarity graph is constructed, which uses the pairwise cosine similarities between all sentence vectors previously computed. In the graph, each node represents a sentence and each edge weight represents its similarity. The PageRank algorithm is then applied to this graph and the sentences are then ranked according to their centrality. The higher the centrality of the sentence, the higher that sentence will be on the list. Finally, the top few sentences are selected as the output[15]. The number of sentences can be defined by the user, however, in this application, the top three sentences are considered. This method is computationally more efficient than abstractive ones and works well on short or medium texts, however, its limitation is that it cannot combine or paraphrase the information resulting in lack of cohesion between the sentences.

Example Input

- Nikola Tesla je bio pronalazač i inženjer.
- Rođen je 1856. godine u Smiljanu.
- Veći deo života proveo je u Sjedinjenim Američkim Državama.

- Poznat je po radu na naizmeničnoj struji.

Translated Input

- Nikola Tesla was an inventor and engineer.
- He was born in 1856 in Smiljan.
- He spent most of his life in the United States.
- He is known for his work on alternating current.

Output

- Nikola Tesla je bio pronalazač i inženjer.
- Rođen je 1856. godine u Smiljanu.

3.2 Abstractive Summarization

Abstractive summarization is a method that takes the original text and tries to shorten it but also keep the essential meaning intact, which requires deeper semantic and contextual understanding of the text[22, 23]. This method was implemented in the project by using the multilingual transformer model `mT5_multilingual_XLSum` , which is based on Text-to-Text Transfer Transformer (T5) architecture, that has proven to be effective in text generation tasks[20]. The model itself was trained on news articles from 45 languages and their corresponding summaries, making it suitable for low-resource languages like Serbian[6]. Since the model shows slightly better accuracy for Cyrillic text, the input is first transliterated accordingly and then processed. The model was tested informally on 30 Serbian snippets, each between 25 and 100 words. It produced fluent Serbian but showed systematic fidelity errors like hallucinated boilerplate in shorter texts, where the model would output warning texts, even though the original text never mentioned this, fabricated facts, over-generalized praise, and truncated outputs on shorter inputs. These errors are consistent with the bias of the XLSum news domain, insufficient Serbian supervision, and the model's tendency to complete a biography.

4 Topic Modelling

Topic modelling is used to automatically extract the main topics present in a collection of texts using Latent Dirichlet Allocation (LDA)[1], which represents each document as a mixture of topics and each topic as a distribution over words. This enables more interpretable representations of the text collection. The input text is normalized, lowercased, tokenized, stripped of punctuation, and filtered using a custom Serbian stopword list before processing. The data is then converted into a document-term matrix using CountVectorizer[16], which counts word frequencies and is compatible with the way LDA models word frequencies. The LDA model was trained using 10 iterations and a fixed random seed for reproducibility, and the number of topics was set between 5 and 10. For each topic, the top 8 most probable words were extracted. The results in the testing of this method showed showed stable behavior across short and longer inputs in the tests.

Example Input

- Učenici i nastavnici su razgovarali o novom školskom projektu, planirali posetu muzeju nauke u Beogradu, pripremali prezentacije o poznatim naučnicima i organizovali radionice kako bi razvili kreativnost i timski duh.

Translated Input

- Students and teachers talked about the new school project, planned a visit to the science museum in Belgrade, prepared presentations about famous scientists, and organized workshops to develop creativity and team spirit.

Output

- školskom - (school)
- učenici - (pupils)
- timski - (team)
- razvili - (developed)
- razgovarali - (talked)
- radionice - (workshops)
- projektu - (project)
- pripremali - (prepared)

5 Visualisation

To further support the interpretation of the linguistic analysis, three additional visualizations were added to the application: a word cloud to show word frequencies, a heatmap for named entity distribution, and a sunburst chart for part-of-speech tags.

5.1 Word Cloud

The word cloud displays the most frequent words in the processed input. The bigger the word in the graph, the higher its frequency[7].

Input for Word Cloud

- Učenici i nastavnici su razgovarali o novom školskom projektu, planirali posetu muzeju nauke u Beogradu, pripremali prezentacije o poznatim naučnicima i organizovali radionice kako bi razvili kreativnost i timski duh.

Figure 3: Word Cloud

5.2 NER Heatmap

The NER heatmap shows how the named entities are distributed across the sentences and entity types. Each cell in the graph represents the number of entities of a given type in a specific sentence, with color intensity indicating the count. This makes it easier to see where specific types of information are concentrated [4]."

Input for NER Heatmap

- Marko i Ana rade u kompaniji Tesla u Beogradu i često putuju u Zagreb.
- Petar je predstavio novi projekat na konferenciji u Parizu i dobio podršku od UNESCO.
- Jelena je završila studije na Univerzitetu u Ljubljani i sada radi u Microsoftu.
- Nikola i Ivana su posetili Rim i Berlin tokom prolećnog odmora.
- Milica se zaposlila u Google-u i preselila se iz Novog Sada u London.

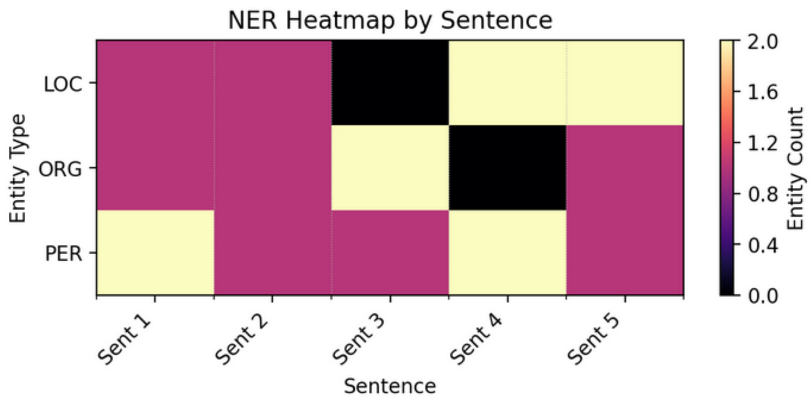

Figure 4: NER Heatmap

5.3 POS Sunburst Chart

The POS Sunburst chart shows the distribution of the part-of-speech tags in a two-level circular structure, where the inner ring shows the universal POS categories and the outer ring shows the corresponding language-specific tags. The size of each segment in the circle reflects the token frequency, enabling a quick overview of the grammatical structure[9].

Input for POS Sunburst Chart

- Učenici i nastavnici su razgovarali o novom školskom projektu, planirali posetu muzeju nauke u Beogradu, pripremali prezentacije o poznatim naučnicima i organizovali radionice kako bi razvili kreativnost i timski duh.

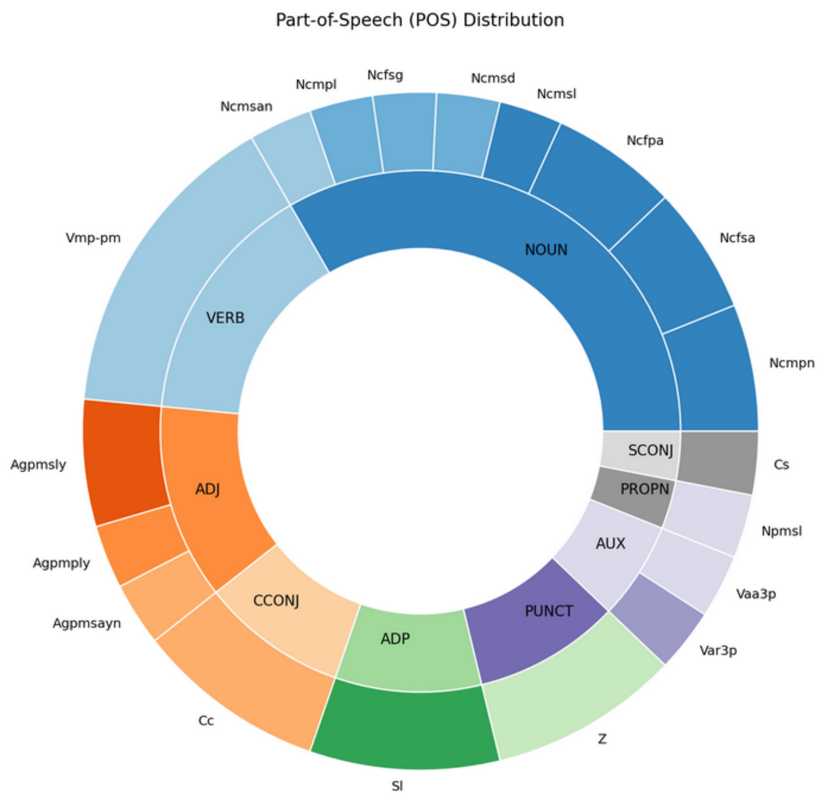

Figure 5: POS Sunburst Chart

6 Appendix

6.1 Repository

The Github repository of the project can be found here. Steps on installation and usage can be found in the README document.

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